

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VII

Roll No.: T009	Name: Siddhesh Kamble
Class: TYIT	Batch: 01
Date of Assignment: 04/08/2026	Date/Time of Submission: 04/08/2026

a. Develop a PWA that works in offline mode using Service Worker.

Code:

offline.html :

```
<!DOCTYPE html>
<html>
<head>
<title>College Portal</title>
<style>
  body {
    font-family: Arial;
    background: #f4f6f9;
    margin: 0;
  }
  .header {
    background: #003366;
    color: white;
    text-align: center;
    padding: 10px;
  }
  .container {
    width: 80%;
    margin: auto;
    margin-top: 20px;
  }

  .card {
    background: white;
    padding: 15px;
    margin-top: 15px;
    border-radius: 8px;
  }
  h3 {
    color: #003366;
  }
  button {
    padding: 8px 15px;
    margin: 5px;
    border: none;
    color: white;
    cursor: pointer;
  }
  .blue {
    background: #007bff;
  }
  .green {
    background: #28a745;
  }
}
```

```

        .red {
            background: #dc3545;
        }
        .offline {
            color: red;
            text-align: center;
        }
    }
</style>
</head>
<body>
    <div class="header">
        <h2>Laxmi Charitable Trust's</h2>
        <h3>Sheth L.U.J College</h3>
        <p>College Portal</p>
    </div>
    <div class="container">
        <div class="card">
            <h2 style="text-align:center;">⚠ You are Offline</h2>
            <p class="offline">No internet connection</p>
        </div>
        <div class="card">
            <h3>Student Details</h3>
            <p>Name: Siddhesh Kamble</p>
            <p>Roll No: T009</p>
            <p>Class: TYIT</p>
            <p>Batch: 01</p>
        </div>
        <div class="card">
            <h3>College Info</h3>
            <p>Department: IT</p>
            <p>University: Mumbai University</p>
            <p>Location: Andheri</p>
        </div>

        <div class="card" style="text-align:center;">
            <h3>Actions</h3>
            <button class="blue">Attendance</button>
            <button class="green">Results</button>
            <button class="blue">Assignments</button>
            <button class="red">Logout</button>
        </div>
    </div>
</body>
</html>

```

service-worker.js :

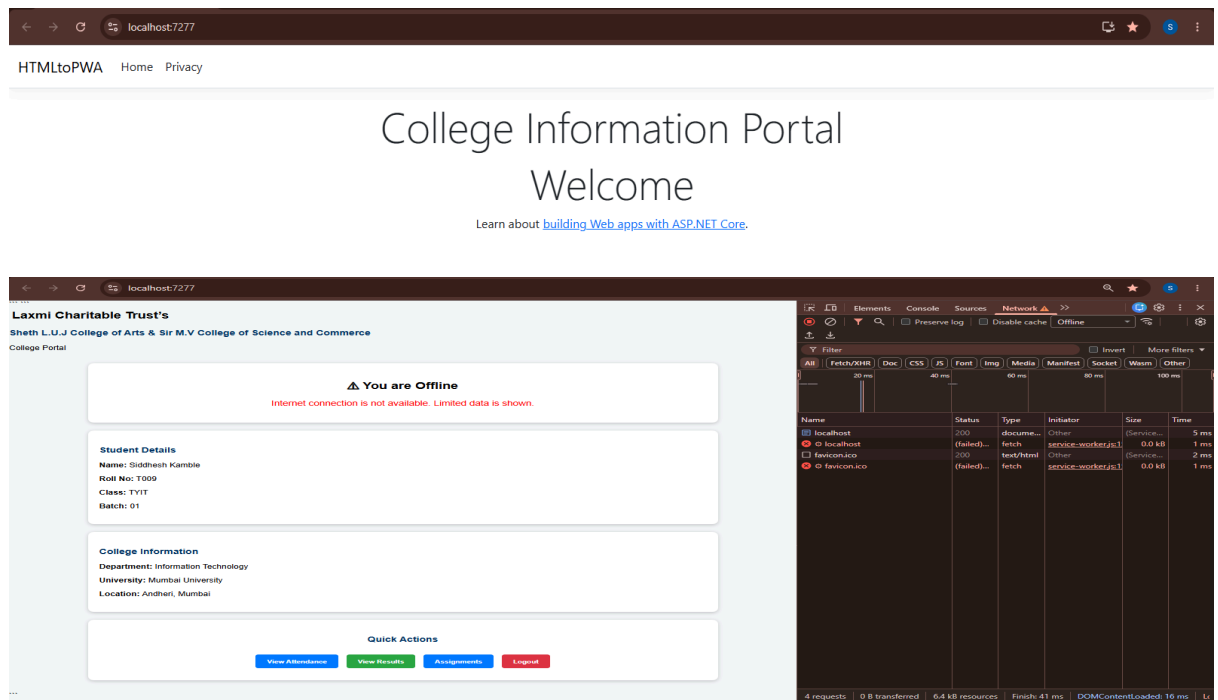
```

const CACHE = "v1";
self.addEventListener("install", event => {
    event.waitUntil(
        caches.open("offline-cache").then(cache => {
            return cache.addAll([
                "/",
                "/offline.html"
            ]);
        });
    );
});

console.log("Service workers Installed ")
});
self.addEventListener("fetch", event => {
    event.respondWith(
        fetch(event.request).catch(() => {
            return caches.match("/offline.html");
        })
    );
});

```

});

Output:**b. Create a Student Record Management PWA.****HomeController :**

```
using Microsoft.AspNetCore.Mvc;
using Practical6.Models;
using System.Diagnostics;
using WebApplication8.Models;
namespace Practical6.Controllers
{
    public class HomeController : Controller
    {
        public IActionResult Index()
        {
            student s = new student();
            s.RollNo = 101;
            s.Name = "Vinayak";
            return View(s);
        }
        public IActionResult Privacy()
        {
            return View();
        }
        [ResponseCache(Duration = 0, Location = ResponseCacheLocation.None, NoStore = true)]
        public IActionResult Error()
        {
            return View(new ErrorViewModel { RequestId = Activity.Current?.Id ?? HttpContext.TraceIdentifier });
        }
    }
}
```

index.cshtml :

```
@{
    ViewData["Title"] = "Home Page";
}
<div class="text-center">
    <h1 class="display-4">College Portal</h1>
    <p>Learn about <a href="https://learn.microsoft.com/aspnet/core">building Web apps with ASP.NET Core</a>.</p>
```

```
<h1>Offline Page</h1>
<h2>Student Record</h2>
<p>Roll No : @Model.RollNo</p>
<p>Name : @Model.Name</p>
</div>
```

Student.cs :

```
namespace WebApplication8.Models
{
    public class student
    {
        public int RollNo { get; set; }
        public string Name { get; set; }
    }
}
```

service-worker.js :

```
const CACHE = "v1";
self.addEventListener("install", e => {
    e.waitUntil(
        caches.open(CACHE).then(cache => {
            return cache.addAll([
                "/",
                "/Home/Index",
                "/css/site.css",
                "/js/site.js"
            ]);
        })
    );
});
self.addEventListener("fetch", e => {
    e.respondWith(
        caches.match(e.request).then(response => {
            return response || fetch(e.request);
        })
    );
});
```

Output :